



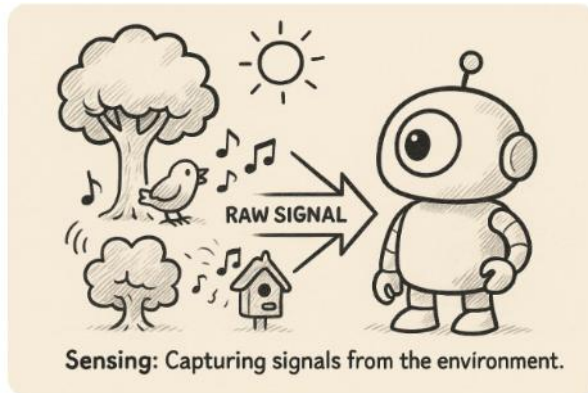
SENSORY SIGNAL TRANSFORMATION

Dr TIAN Jing

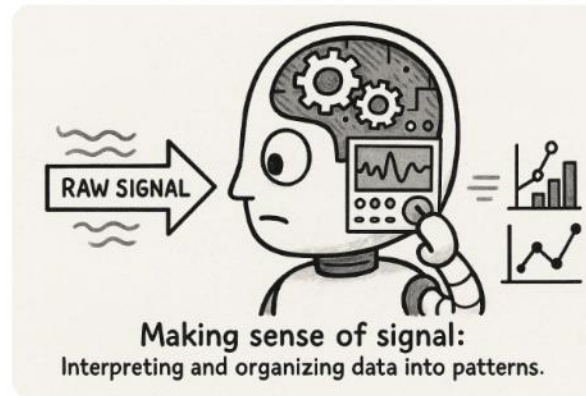
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Recap: Roadmap

Sensing: Capture raw data from the environment using sensors.



Making sense of signal: Transform and interpret the raw signal into meaningful patterns or features.



Sense making: Use interpreted signals to derive understanding or decision-making.

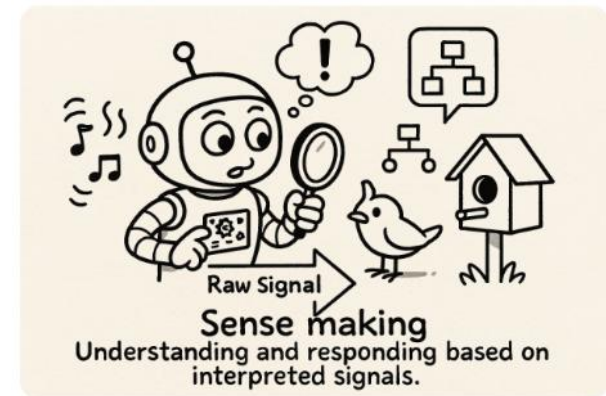


Image source: GPT

Physical Domain (our focus)

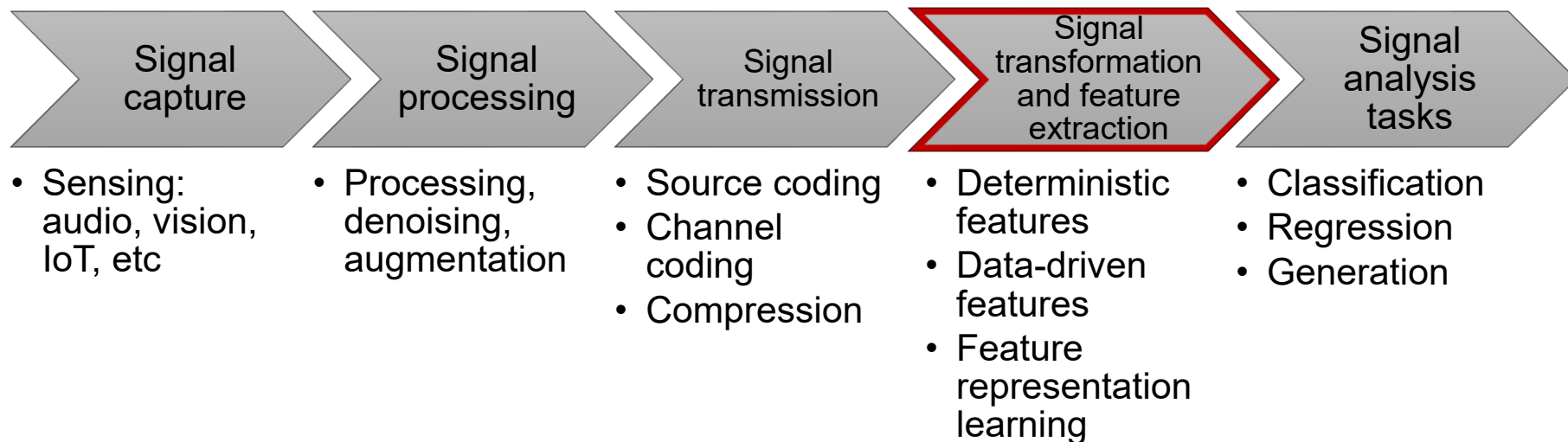
- Sensor-driven systems, e.g. IoT

Business Domain

- Non-sensor data, e.g., transactions, customer data
- Also need sensor data (e.g., customer's location)



Signal processing pipeline

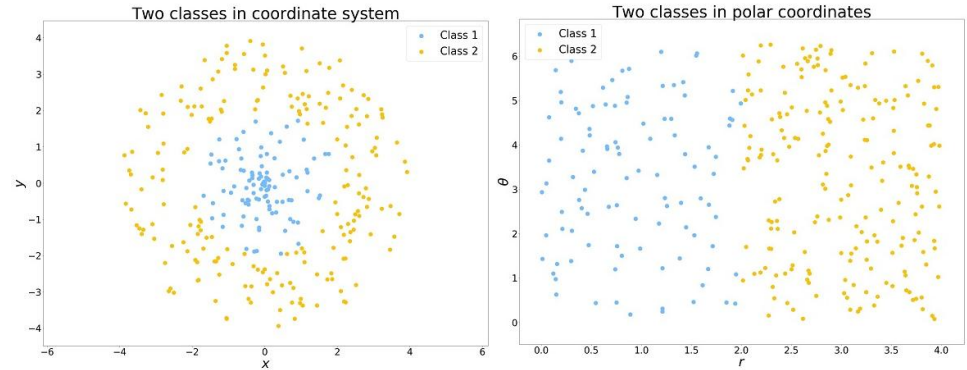


Signal representation/Feature extraction (similar to feature engineering) uses mathematical transformations of raw input data to create new features to be used for further machine learning models.

Feature representation for sensory data

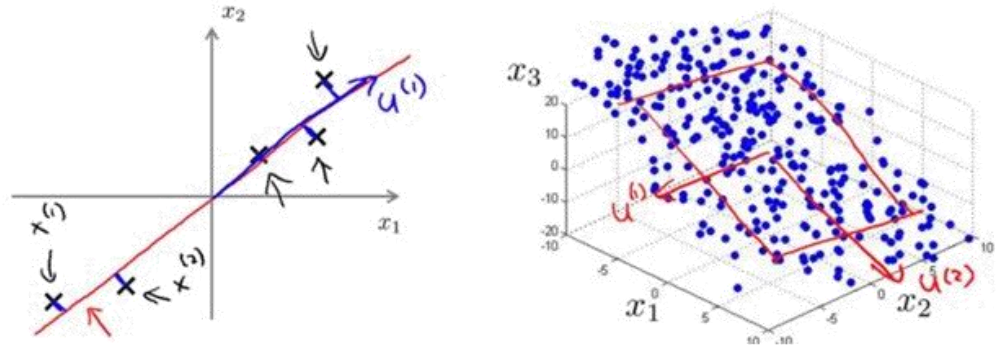
1. Signal representation can be **manually designed** (input-agnostic)

[Covered in today's class]



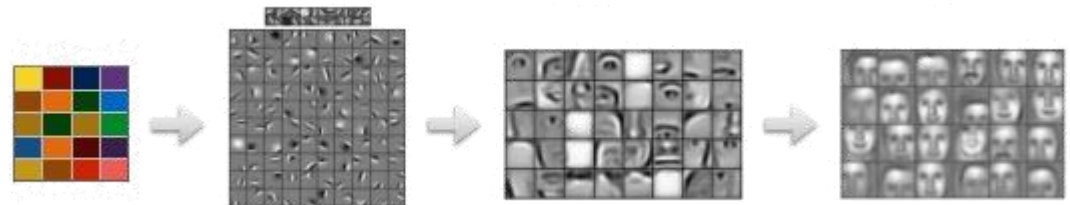
2. Signal representation can be **adaptively to the input signal**

[Covered in tomorrow's class]



3. Signal representation can be **learned from signal dataset**

[Covered in tomorrow's class]





FOUNDATIONS OF SENSORY SIGNAL PROCESSING

FREQUENCY DOMAIN



Time-frequency analysis of signal

One of the top 10 algorithms in the 20th century!

1. Metropolis algorithm for Monte Carlo
2. Simplex method for linear programming
3. Krylov subspace iteration
4. Decomposition approach to matrix computation (Singular value)
5. The Fortran compiler
6. QR algorithm for eigenvalues
7. Quick sort
- 8. Fast Fourier transform (FFT)**
9. Integer relation detection
10. Fast multipole

Fourier Compressor: Frequency-Domain Visual Token Compression for Vision-Language Models

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⁴ School of Computer Science, Shanghai Jiao Tong University

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🔗 <https://github.com/whyisverysmart/Fourier-Compressor>

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LATENT FOURIER TRANSFORM

ICLR 2026

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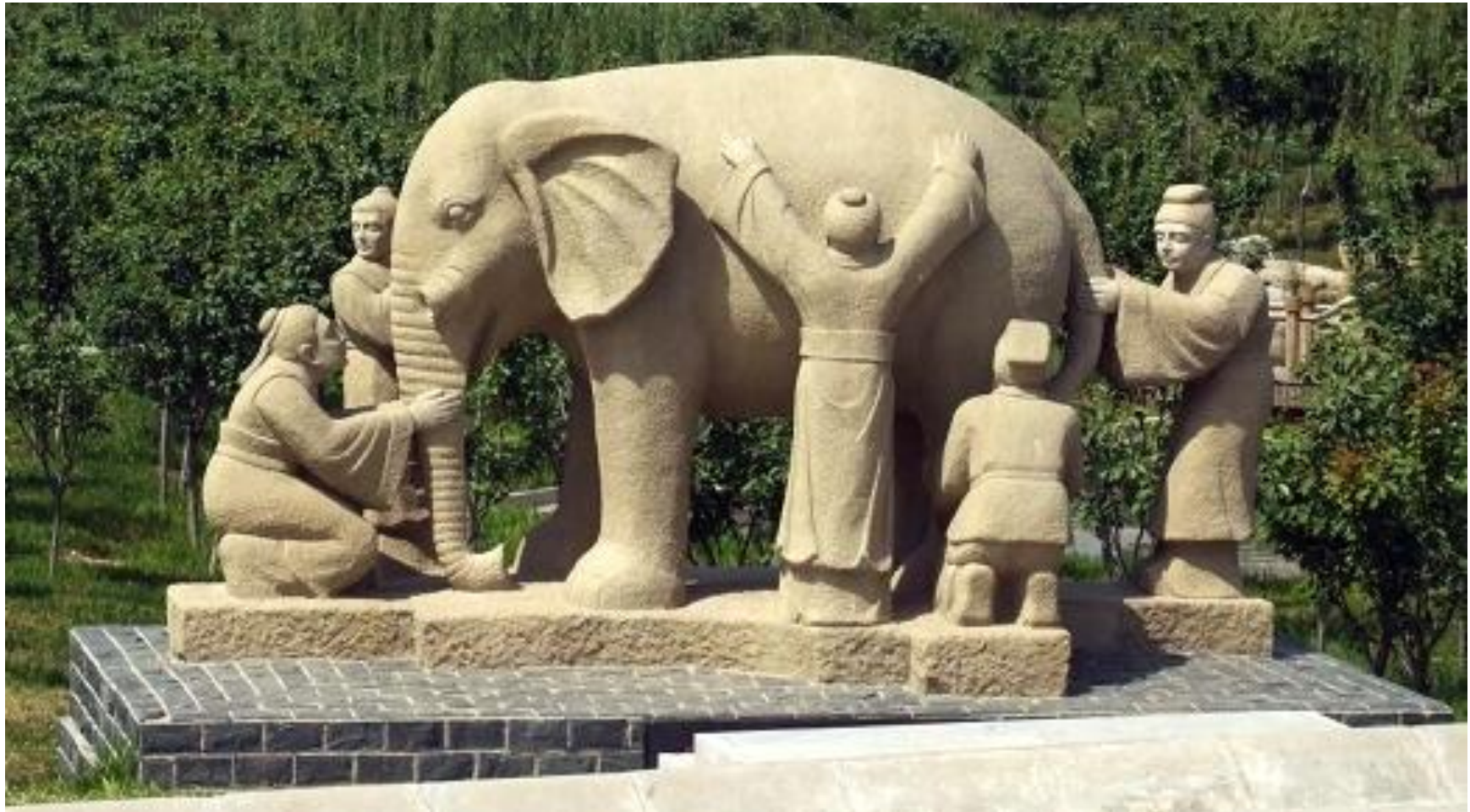
huangcza@mit.edu

Source: <https://www.computer.org/csdl/mags/cs/2000/01/c1022.html>



Signal decomposition (intuition)

Can you describe the animal object in this image (without mentioning its name) so that a listener can visualize what you are describing?



Source: <https://cellcode.us/quotes/and-parable-blind-men-elephant.html>

Signal decomposition (image)

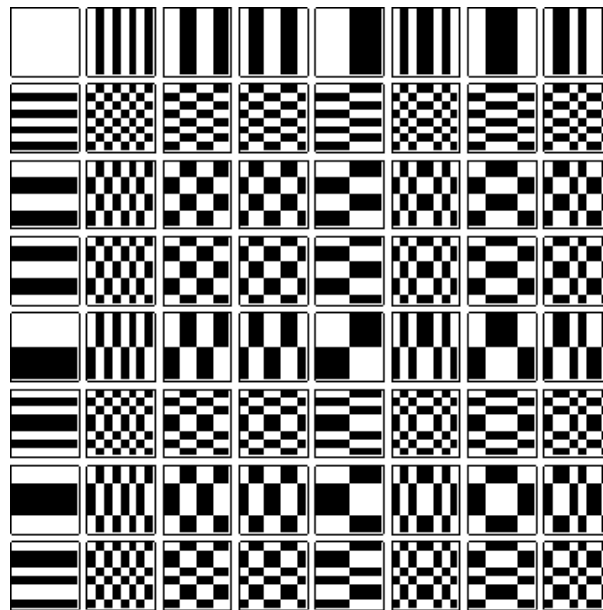
Objective: Decompose signal into “basis”, which is determined by certain frequency.

- Image signal: Checkerboard basis.
- Images have some slow/fast varying regions. For example, a first checkerboard picture with (slow-varying) constant color. A second checkerboard picture that has fast changes.

Input image



A set of basis (checkerboard)

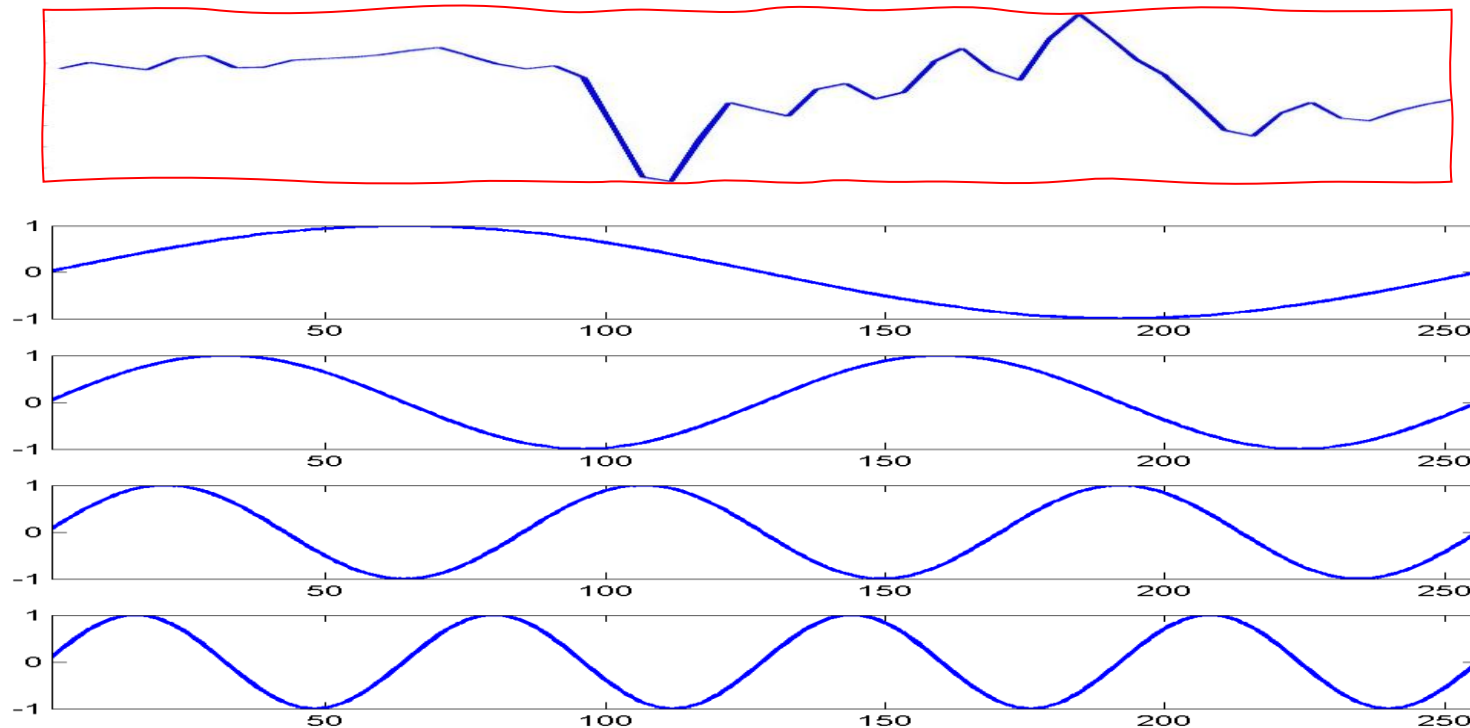


Note: The images are NOT visualized in the actual scale.

Signal decomposition (sound)

Objective: Decompose signal into “basis”, which is determined by certain frequency.

- Sound signal: Sinusoidal basis.
- They are orthogonal.



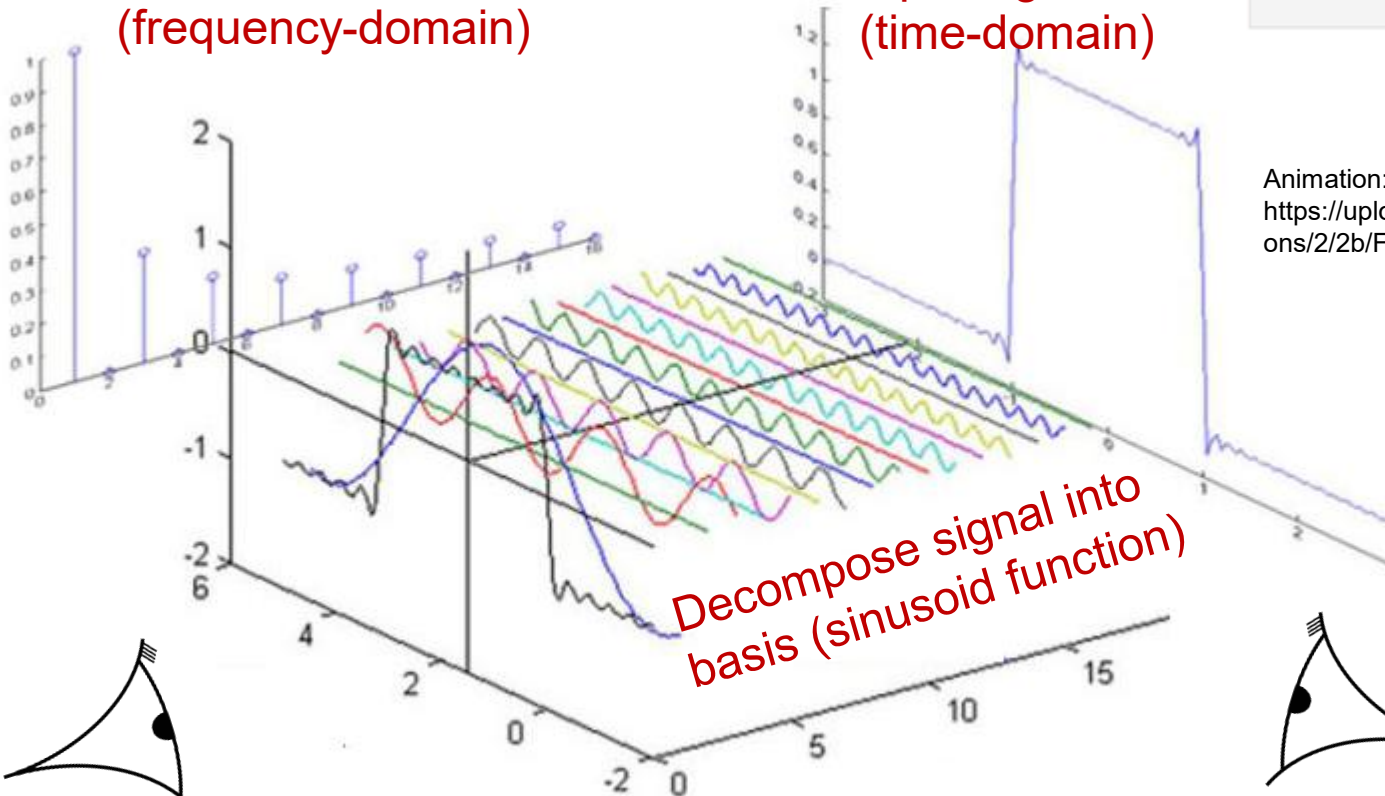
Signal decomposition (idea)

Idea: Decompose the input signal from the time domain into a set of basis (sinusoidal functions) to obtain the feature (measurement) in the frequency domain.



Feature (measurement)
(frequency-domain)

Input signal
(time-domain)



Animation:
https://upload.wikimedia.org/wikipedia/commons/2/2b/Fourier_series_and_transform.gif



View for time-domain



View for frequency-domain



Signal decomposition (Sinusoids basis)

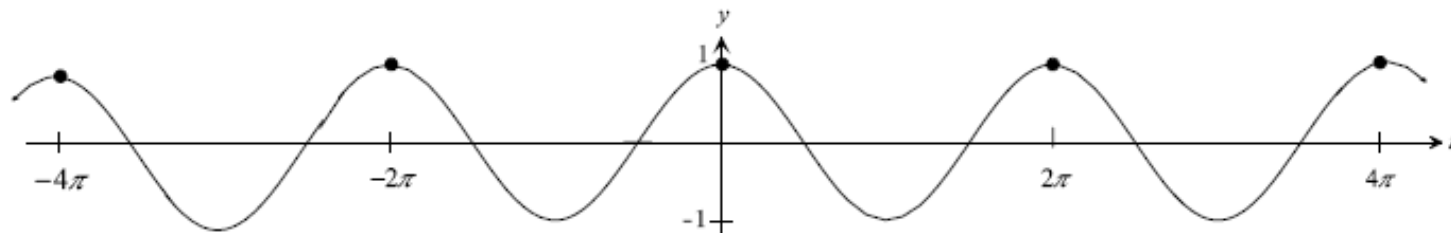
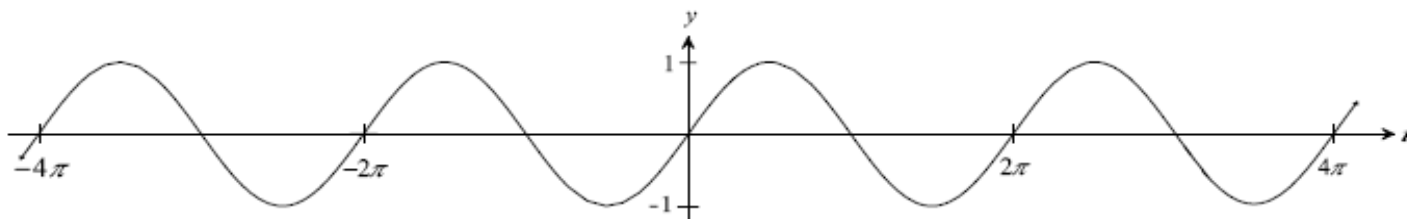
- Periodic functions
- General form of sine and cosine functions

$ A $	amplitude
$\frac{2\pi}{ \alpha }$	period
b	phase shift

$$y(t) = A \sin(\alpha t + b)$$

$$y(t) = A \cos(\alpha t + b)$$

Example: $A = 1, b = 0, \alpha = 1$ period = 2π





Signal decomposition (Fourier analysis)

Signal
(Fourier domain)

Apply all
signal values

Signal
(time domain)

Basis
(sinusoidal functions)

$$\text{Forward: } F(u) = \sum_{x=0}^{N-1} f(x) e^{\frac{-i2\pi ux}{N}}, \text{ where } u = 0, 1, \dots, N-1$$

$$\text{Inverse: } f(x) = \frac{1}{N} \sum_{u=0}^{N-1} F(u) e^{\frac{i2\pi ux}{N}}, \text{ where } x = 0, 1, \dots, N-1$$

Note

$e^{ix} = \cos x + i \sin x$; $e^{i\pi} = \cos \pi + i \sin \pi = -1$,

Reference:

https://en.wikipedia.org/wiki/Euler%27s_identity

Example

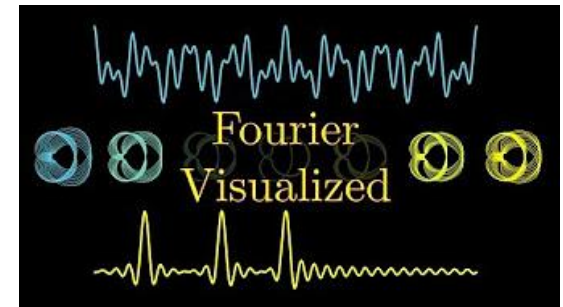
- Signal: $f(x) = [2, 3, 4, 4]$
- Fourier coefficients: $F(u) = [13, (-2 + i), -1, (-2 - i)]$, where i is the imaginary unit

$$F(0) = \sum_{x=0}^3 f(x) e^{\frac{-i2\pi 0x}{4}} = 2 + 3 + 4 + 4 = 13$$

$$F(1) = \sum_{x=0}^3 f(x) e^{\frac{-i2\pi x}{4}} = 2e^0 + 3e^{-i\pi/2} + 4e^{-i\pi} + 4e^{-i3\pi/2} = -2 + i$$

$$F(2) = \sum_{x=0}^3 f(x) e^{\frac{-i4\pi x}{4}} = 2e^0 + 3e^{-i\pi} + 4e^{-i2\pi} + 4e^{-i3\pi} = -1$$

$$F(3) = \sum_{x=0}^3 f(x) e^{\frac{-i6\pi x}{4}} = 2e^0 + 3e^{-i3\pi/2} + 4e^{-i3\pi} + 4e^{-i9\pi/2} = -2 - i$$



What is the Fourier Transform? A visual introduction

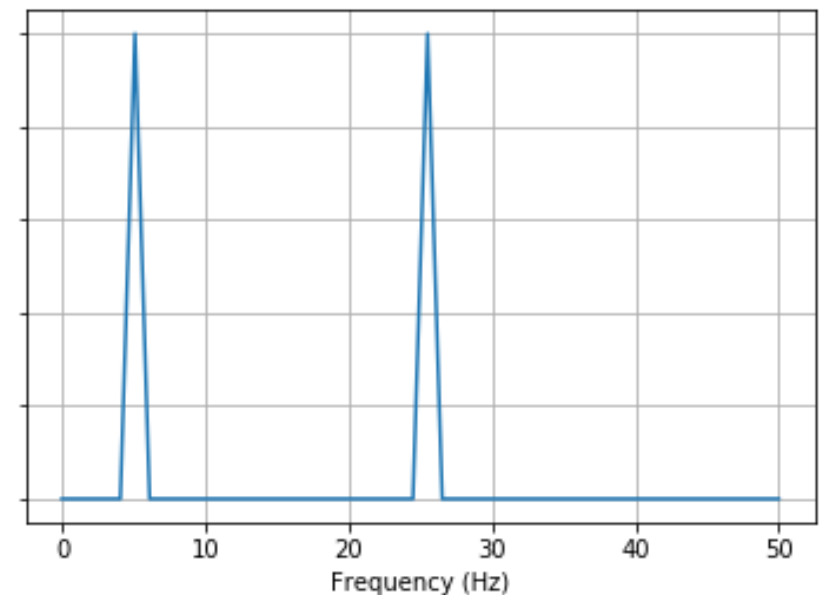
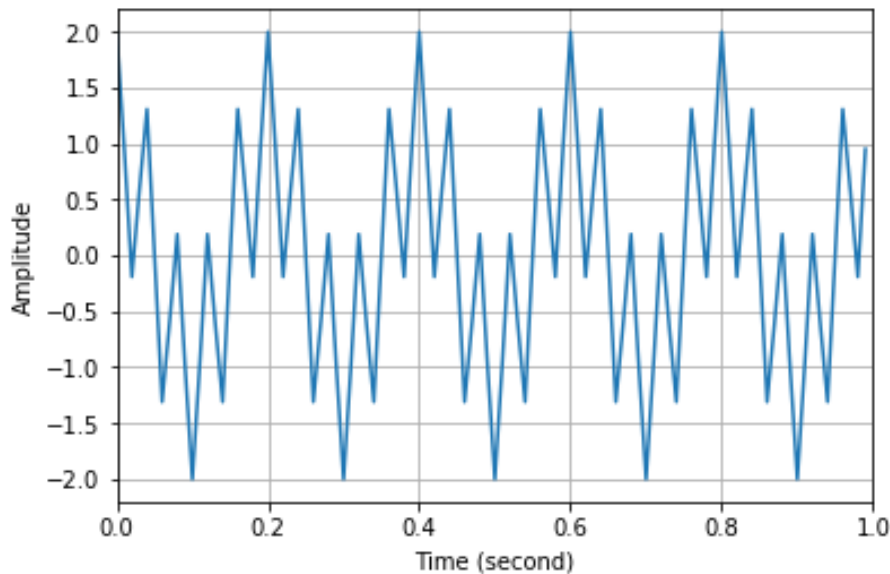
A 20-minute video tutorial on

<https://www.youtube.com/watch?v=spUNpyF58BY>

12 million views since January 2018

Challenge: Provides good localization in the frequency domain but poor localization in the time domain. Knows what frequencies exist (example 1 in the current slide), but no information about where these frequencies are in time (two examples in the next slide).

Example 1: $f(t) = \cos(2\pi \cdot 5 \cdot t) + \cos(2\pi \cdot 25 \cdot t)$

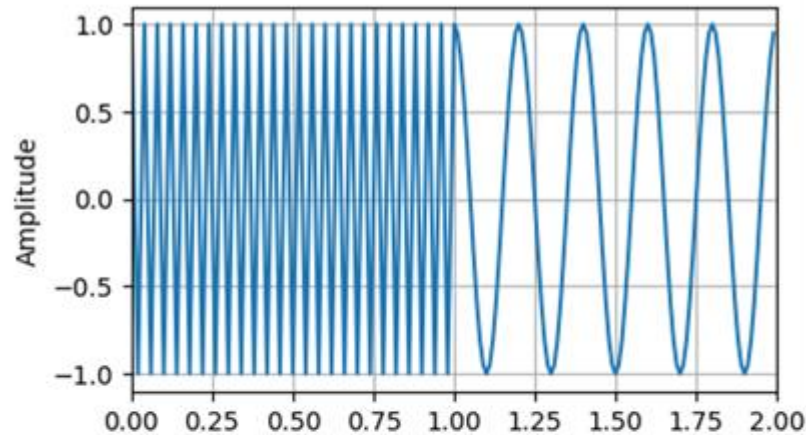


Fourier analysis

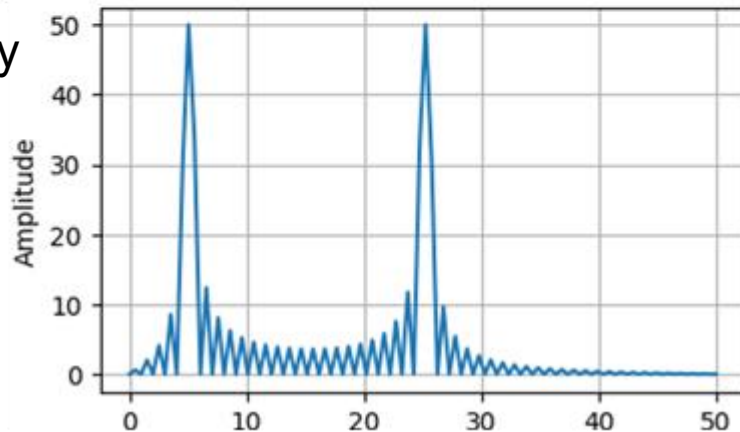
Example 2:

$$f(t) = \begin{cases} \cos(2\pi \cdot 25 \cdot t) & 0 \leq t < 1 \\ \cos(2\pi \cdot 5 \cdot t) & 1 \leq t < 2 \end{cases}$$

Time domain

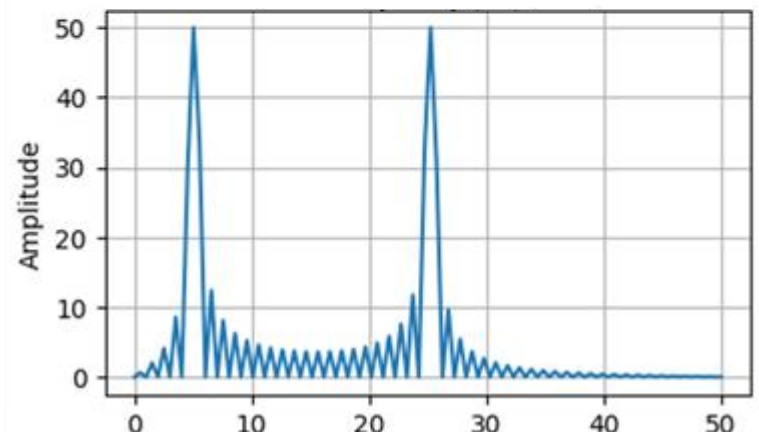
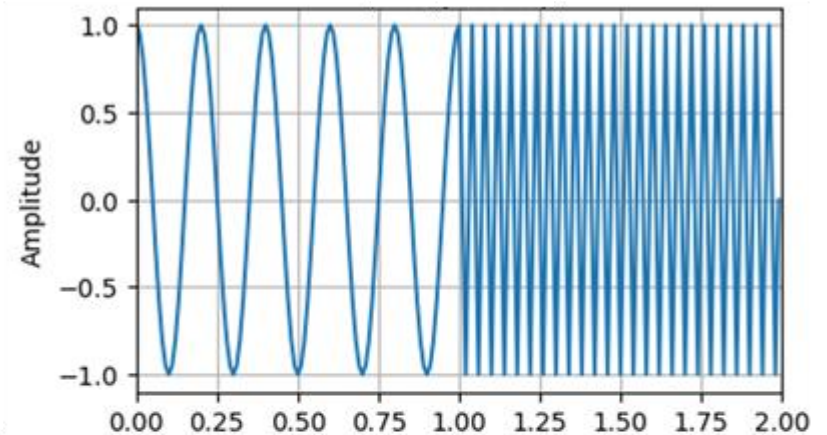


Frequency domain



Example 3:

$$f(t) = \begin{cases} \cos(2\pi \cdot 5 \cdot t) & 0 \leq t < 1 \\ \cos(2\pi \cdot 25 \cdot t) & 1 \leq t < 2 \end{cases}$$

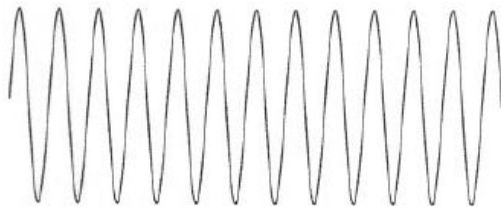




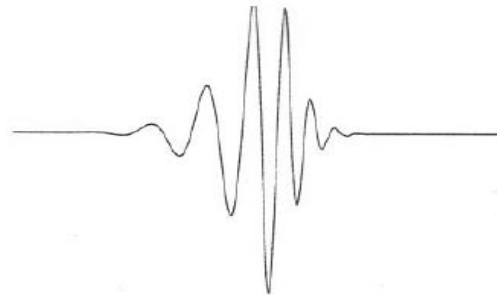
What is a wavelet?

- A function that “waves” above and below the x -axis with the following properties
 - Varying frequency
 - Limited duration
- This is in contrast to sinusoids, used by the Fourier transform, which have a constant frequency.

Sinusoid



Wavelet



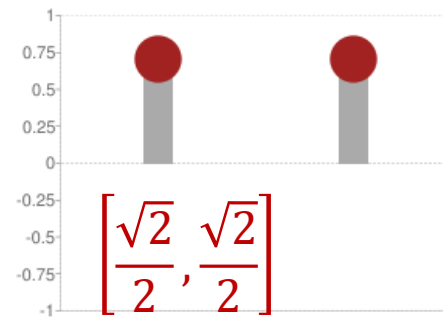
Wavelet transform (overview)



- **Decompose** signal recursively by using a set of decomposition filters (including hand-crafted low-pass filters and high-pass filters) to obtain wavelet coefficients, which are grouped in subbands (different decomposition levels/filters).
- **Reconstruct** signal recursively by using a set of reconstruction filters (including hand-crafted low-pass filters and high-pass filters).
- Low-pass filter: Smoothing, averaging. High-pass filter: Differencing.
- There are many types of filters (see <http://wavelets.pybytes.com/wavelet/haar/>).

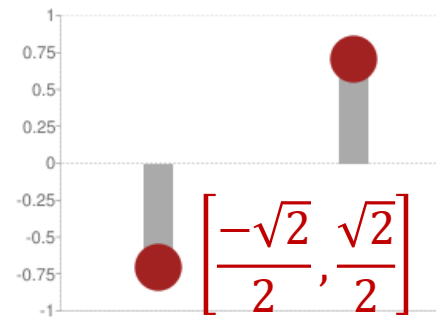
Example: Haar wavelet

Decomposition low-pass filter (copy!)



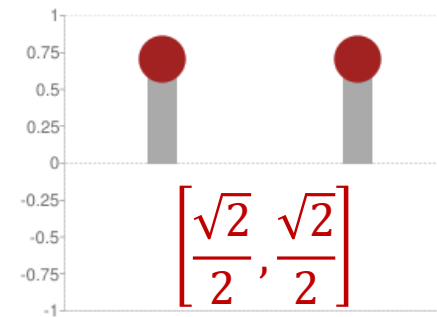
$h_0 = 0.7071067812$
 $h_1 = 0.7071067812$

Decomposition high-pass filter (copy!)



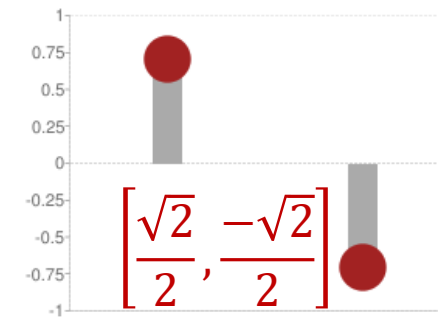
$g_0 = -0.7071067812$
 $g_1 = 0.7071067812$

Reconstruction low-pass filter (copy!)



$h'_0 = 0.7071067812$
 $h'_1 = 0.7071067812$

Reconstruction high-pass filter (copy!)



$g'_0 = 0.7071067812$
 $g'_1 = -0.7071067812$



Wavelet decomposition (example)

- Decompose signal using a low-pass filter and a high-pass filter to get approximation coefficients and detail coefficients, followed by down-sampling.

Low-pass filter $\left[\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right]$
with zero extension

$$(9 * \frac{\sqrt{2}}{2} + 7 * \frac{\sqrt{2}}{2}) = 8\sqrt{2}$$

$$[8\sqrt{2} \quad 5\sqrt{2}]$$

Down-sampling ratio 2

Low-pass filter $\left[\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right]$
with zero extension

Down-sampling ratio 2

Signal: [9 7 3 5]

High-pass filter $\left[\frac{-\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right]$
with zero extension

$$\left(9 * \frac{-\sqrt{2}}{2} + 7 * \frac{\sqrt{2}}{2}\right) = -\sqrt{2}$$

$$[-\sqrt{2} \quad -2\sqrt{2}]$$

Down-sampling ratio 2

High-pass filter $\left[\frac{-\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right]$
with zero extension

Down-sampling ratio 2

$$[8\sqrt{2} \quad 4\sqrt{2}]$$

$$[-\sqrt{2} \quad \sqrt{2}]$$

$$[12 \quad 4]$$

$$[-4 \quad -4]$$

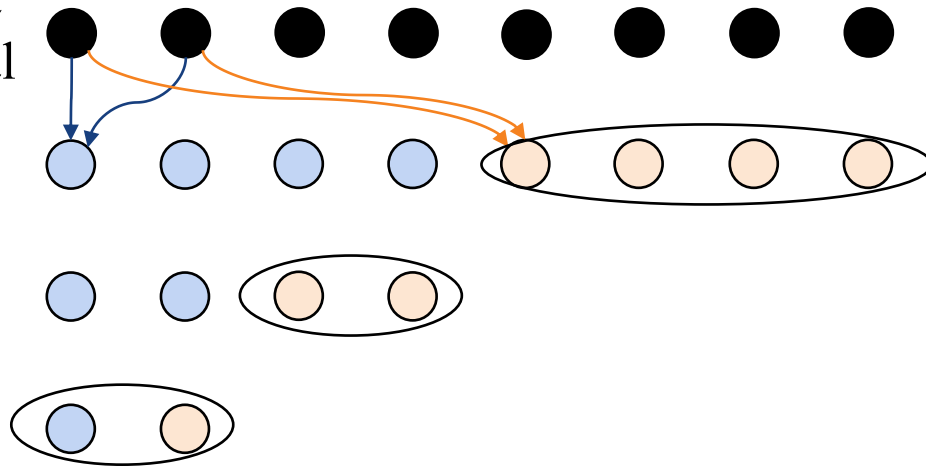
$$[12]$$

$$[-4]$$

Wavelet coefficients: [12 -4 $-\sqrt{2}$ $\sqrt{2}$]

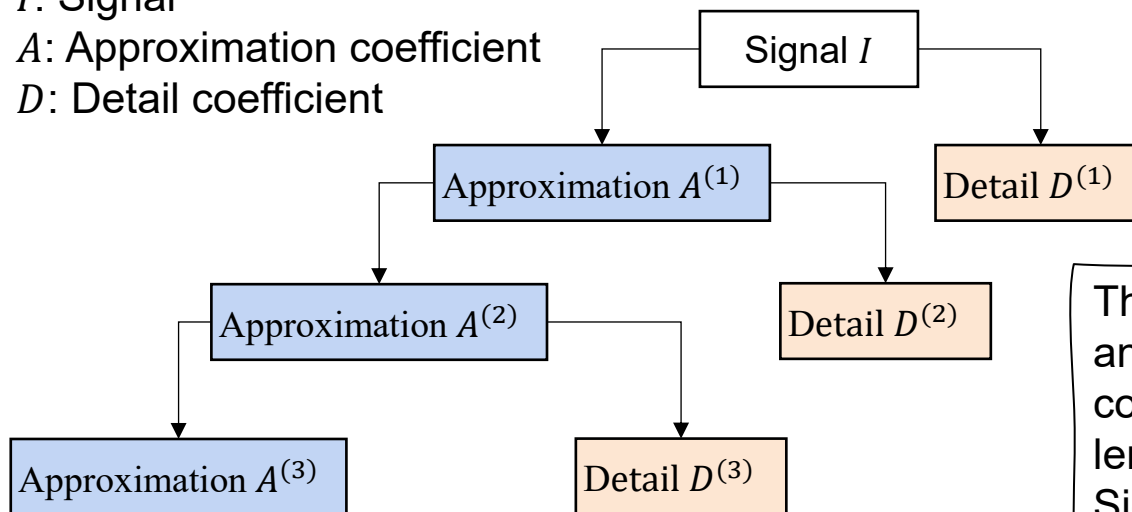
Wavelet decomposition (overview)

Input
signal



- Approximation
(low-pass filter, down-sampling)
- Detail
(high-pass filter, down-sampling)

- I : Signal
- A : Approximation coefficient
- D : Detail coefficient



The maximum decomposition level and the dimension of output coefficients depend on (i) Signal length, (ii) Filter length, and (iii) Signal extension method.



Wavelet reconstruction (example)

Reconstruct the signal using a low-pass filter and a high-pass filter.

Elementwise add

Reconstructed signal: [9 7 3 5]

Low-pass filter $\left[\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right]$
with zero extension

High-pass filter $\left[\frac{\sqrt{2}}{2}, \frac{-\sqrt{2}}{2}\right]$
with zero extension

Up-sampling ratio 2

Up-sampling ratio 2

Elementwise add

Low-pass filter $\left[\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right]$
with zero extension

High-pass filter $\left[\frac{\sqrt{2}}{2}, \frac{-\sqrt{2}}{2}\right]$
with zero extension

Up-sampling ratio 2

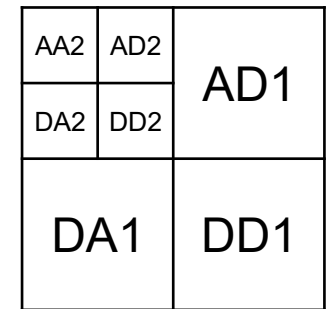
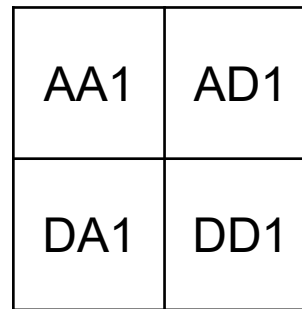
Up-sampling ratio 2

Wavelet coefficients: [12 -4 $-\sqrt{2}$ $\sqrt{2}$]

Note: It is okay that we insert 0s at even positions in up-sampling. That will cause the reconstructed signal shifted along the time axis.

Appendix: 2D wavelet transformation

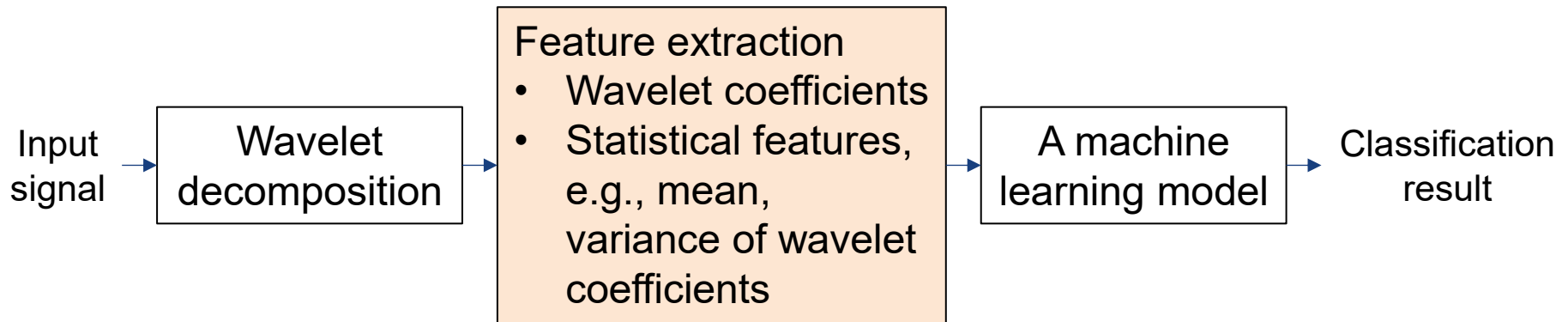
- **AA**: The upper left quadrant is filtered by the low-pass filter along the rows and then filtered along the corresponding columns with the low-pass filter. It represents the approximation coefficients of the original at half the resolution.
- **DA/AD**: The lower left and the upper right blocks are filtered along the rows and columns with a low-pass filter and a high-pass filter, alternatively. The AD block contains vertical edges. In contrast, the HL blocks show horizontal edges.
- **DD**: The lower right quadrant is derived analogously to the upper left quadrant, but with the use of the high-pass filter, where we find edges of the original image in a diagonal direction.



A: Approximation via low-pass filter
D: Detail via high-pass filter

Decomposed into more levels

Application: Feature extraction



Idea: Use wavelet decomposition as a hand-crafted feature extraction

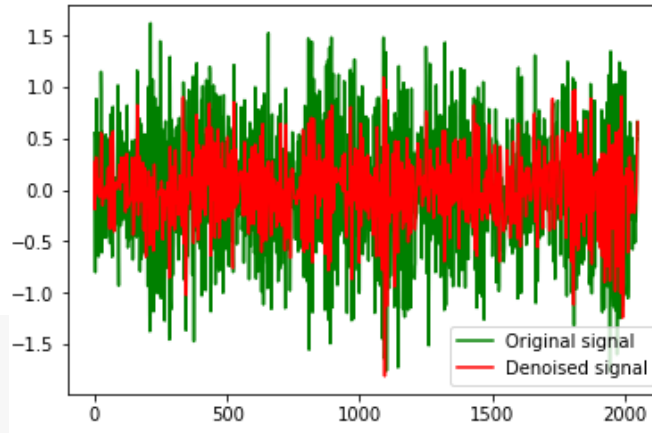
- Perform wavelet transformation on the input signal
- Use wavelet coefficients as features or further extract statistical features like variance, mean, median, etc.
- Perform signal classification or other machine learning tasks

Reference: Image classification using convolutional neural network with wavelet domain inputs, <https://doi.org/10.1049/ipr2.12466>



Application: Signal enhancement

An example of **signal denoising**



```
waveletname = 'db4'
waveletlevel = 2

# Step 1: Perform wavelet decomposition

coeffs_orig = pywt.wavedec(signal_orig, waveletname, level=waveletlevel)
coeffs_filter = coeffs_orig.copy()

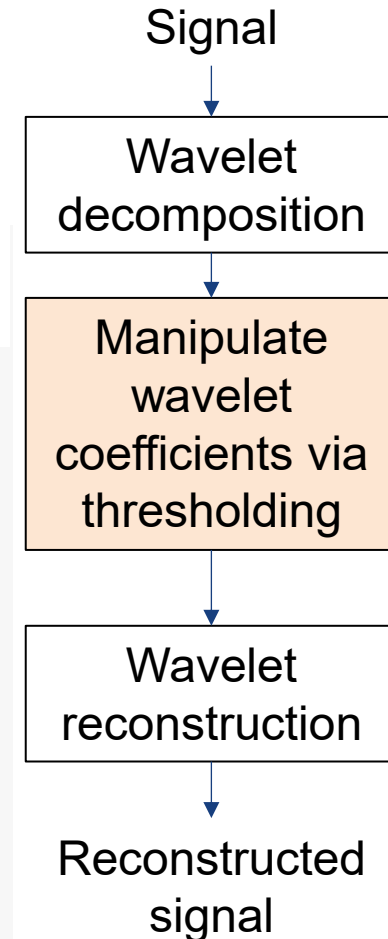
# Step 2: Perform thresholding on the wavelet coefficients

# Set the threshold
threshold = 0.8

for i in range(1, len(coeffs_orig)):
    coeffs_filter[i] = pywt.threshold(coeffs_orig[i], threshold*max(coeffs_orig[i]))

# Step 3: Perform reconstruction on the filtered coefficients

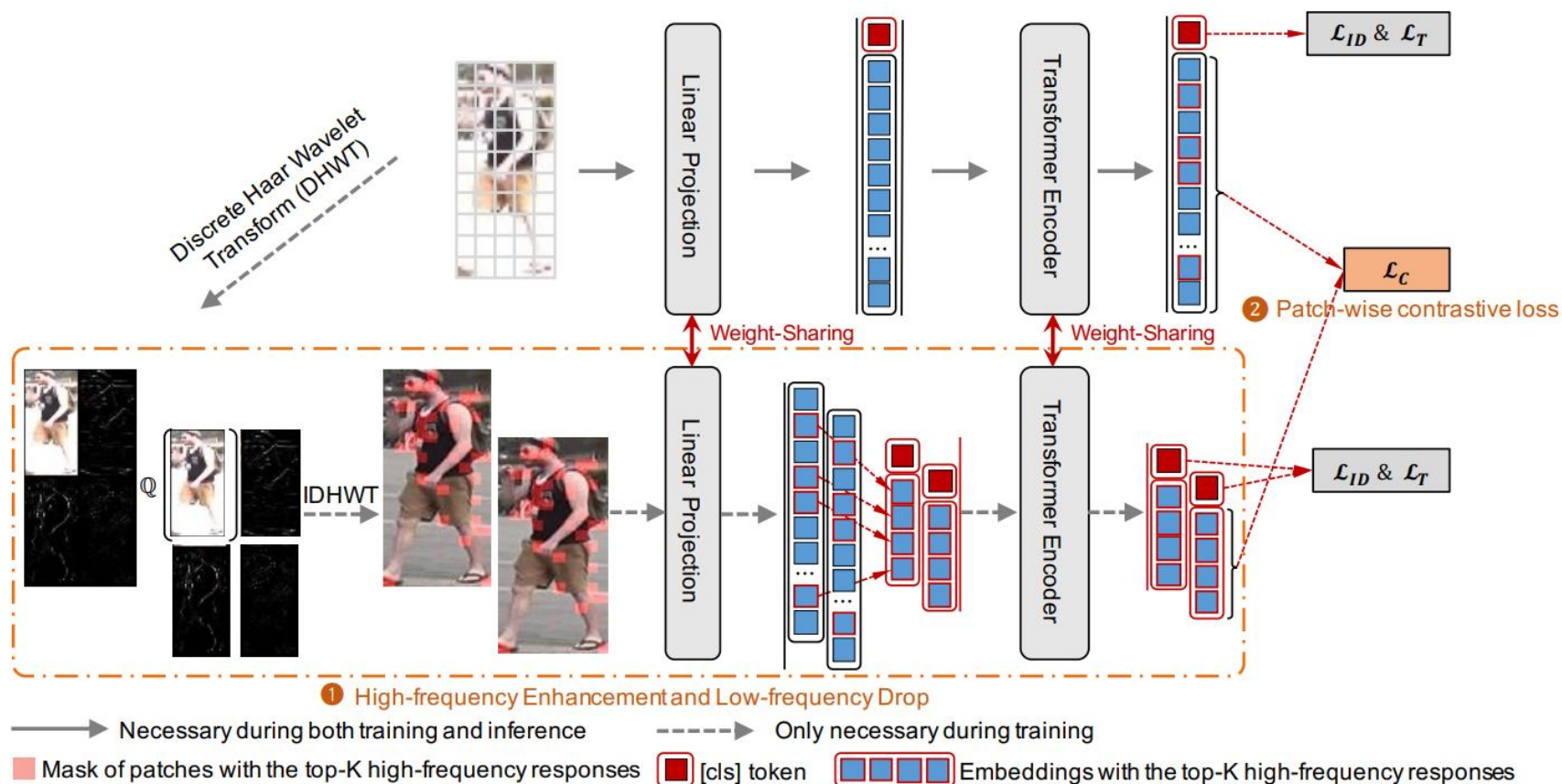
signal_denoised = pywt.waverec(coeffs_filter, waveletname)
```





Application: Data augmentation

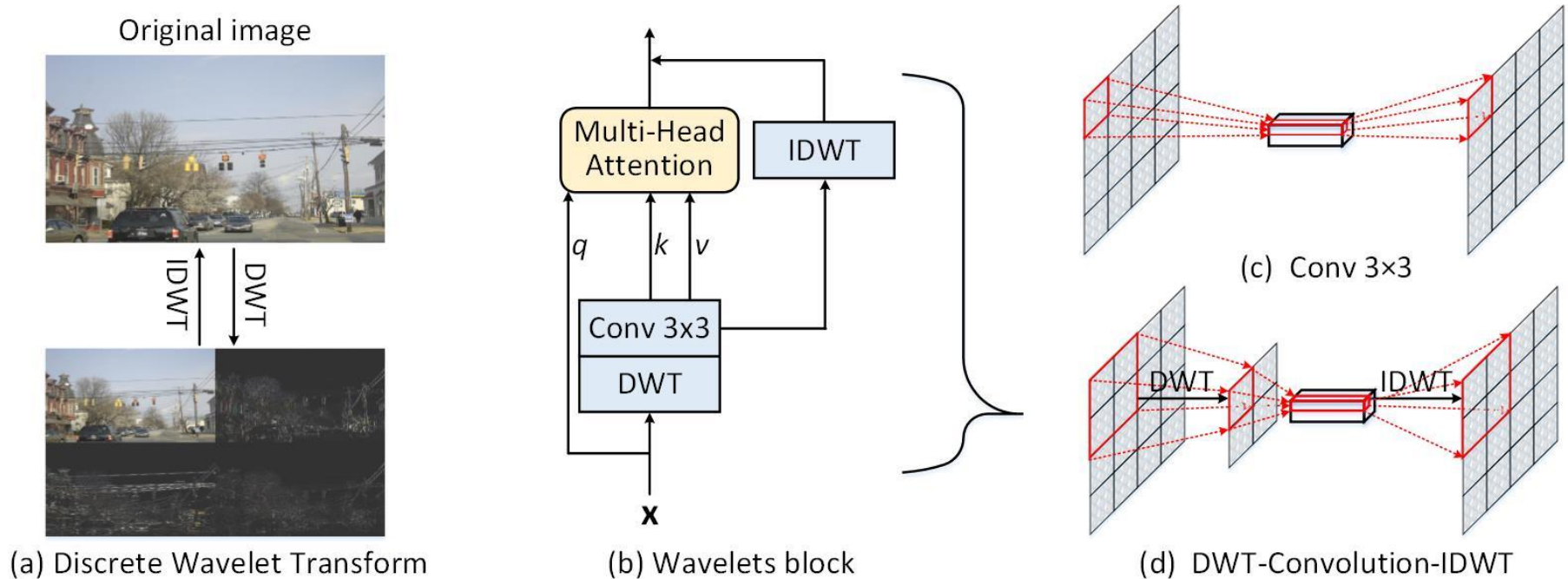
Idea: Manipulate data in the wavelet domain as an augmentation.



Reference: PHA: Patch-wise High-frequency Augmentation for Transformer-based Person Re-identification, CVPR 2023, <https://ieeexplore.ieee.org/document/10205407/>

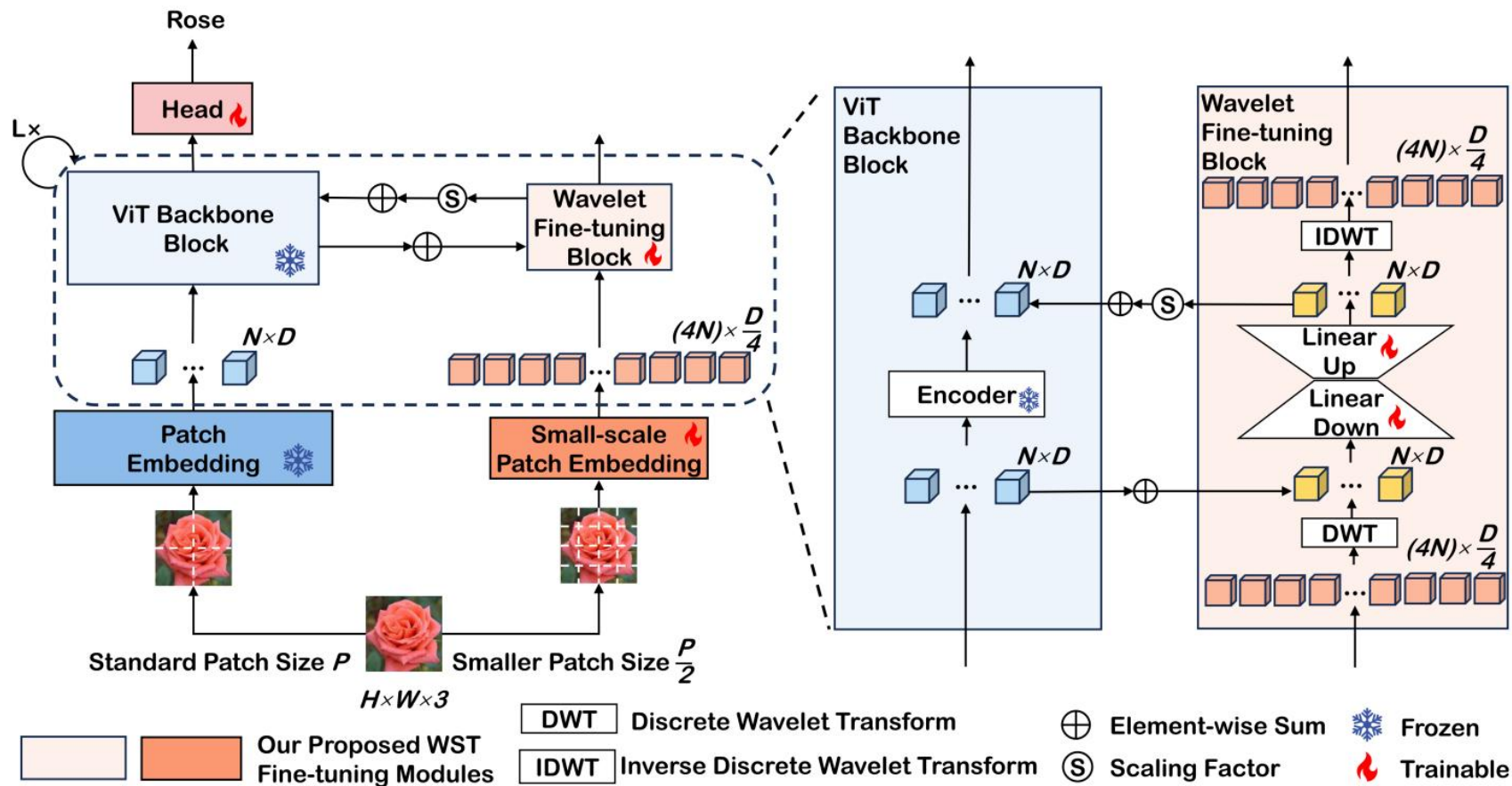
Application: Integration with CNN

Insert wavelet as freezing conv filters



Reference: Wave-ViT: Unifying wavelet and transformers for visual representation learning, ECCV 2022, <https://arxiv.org/abs/2207.04978>

Application: Efficient tuning



Reference: WST: Wavelet-Based Multi-scale Tuning for Visual Transfer Learning, AAAI 2025, <https://ojs.aaai.org/index.php/AAAI/article/view/34387>

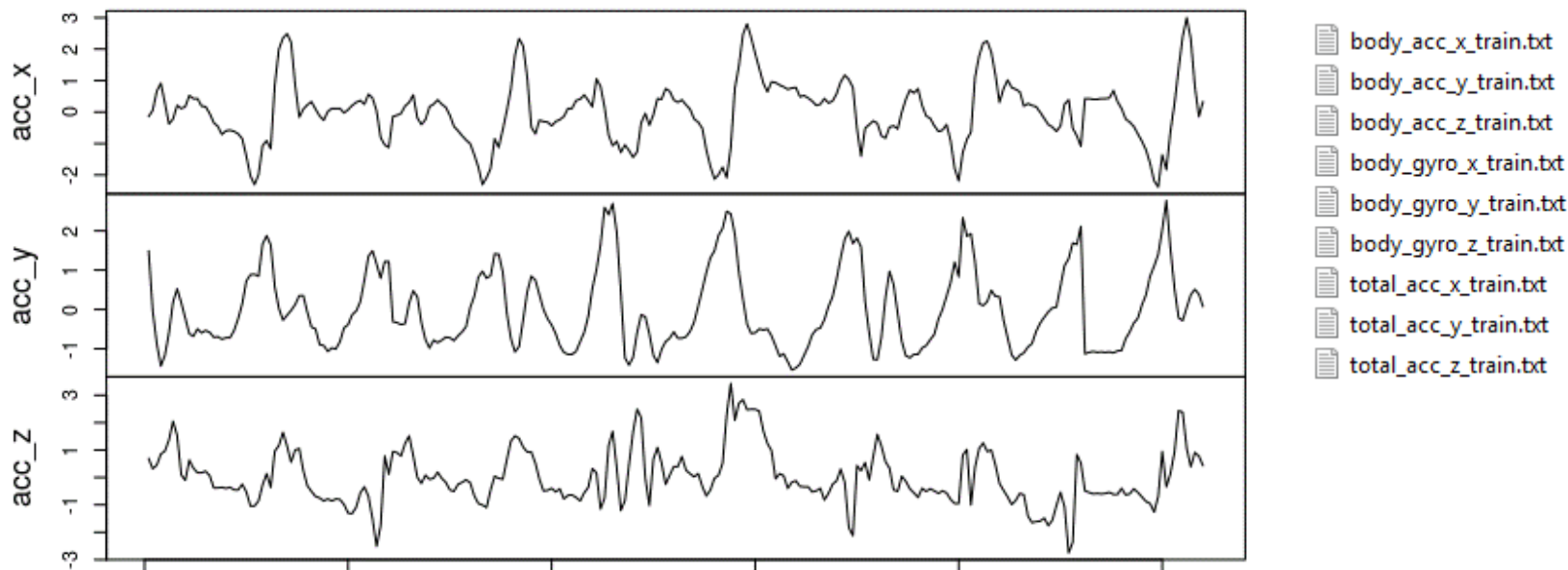


STATISTICAL SENSORY SIGNAL ANALYSIS

Example used in workshop

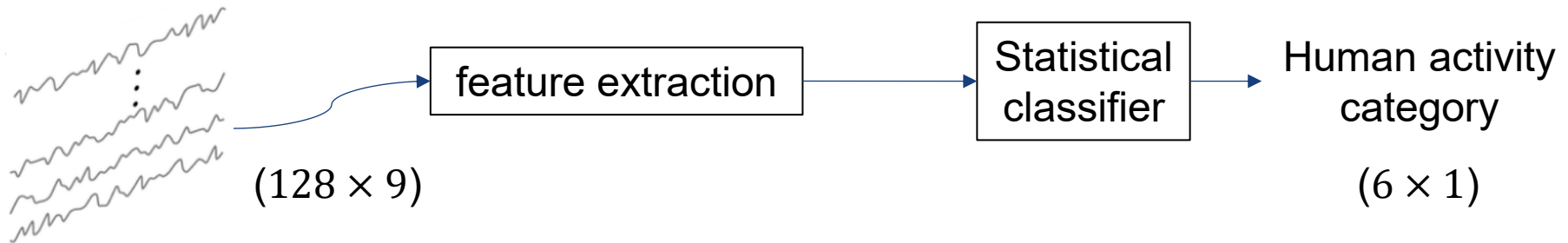
Human Activity Recognition Using Smartphones Data Set

- Each person performed six activities (WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS, SITTING, STANDING, LAYING) wearing a smartphone (Samsung Galaxy S II) on their waist. Using its embedded accelerometer and gyroscope, we captured 3-axial linear acceleration and 3-axial angular velocity at a constant rate of 50Hz.
- Web: <https://archive.ics.uci.edu/ml/datasets/Human+Activity+Recognition+Using+Smartphone>



Example used in workshop

Model input	Model	Model output
Human activity smart phone data	To build a feature extraction method + statistical classification	Human activity (6 categories for this dataset)



Time series
measurements

# of measurements (2.56 seconds, 50 Hz)	128
# of component (X, Y, Z , for total acceleration, body acceleration, body angular velocity, respectively)	9
# of activity	6

A signal classification using a statistical approach (Naïve Bayesian classification).

- Given an observation x , two categories of data, say ω_1, ω_2 , does the signal x belong to ω_1 or ω_2 ?
- That means, we need to evaluate $P(\omega_1|x)$ and $P(\omega_2|x)$ and make decision.
- The class posterior probability is

$$P(\omega_i|x) = \frac{\overset{\text{Likelihood}}{P(x|\omega_i)} \overset{\text{Prior}}{P(\omega_i)}}{\underset{\text{Evidence}}{P(x)}}$$

- **Class prior** $P(\omega_i)$
 - A learning task to formulate this distribution
- **Class likelihood** $P(x|\omega_i)$
 - Requires that we have a model for each ω_i , for example, ω_i can be modelled as a Gaussian distribution
- **Evidence**
 - $P(x) = P(x|\omega_1)P(\omega_1) + P(x|\omega_2)P(\omega_2)$
- **Classification rule**
 - If $P(\omega_1|x) \geq P(\omega_2|x)$, then x is classified as ω_1
 - If $P(\omega_1|x) < P(\omega_2|x)$, then x is classified as ω_2

Categorical data

Given: $x=(\text{Outlook}=\text{Sunny}, \text{Temperature}=\text{Cool}, \text{Humidity}=\text{High}, \text{Wind}=\text{Strong})$
Predict: PlayTennis Yes or No?

PlayTennis: training examples

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Note: It is assumed that the variables that are used for the prediction are independent of each other.



Categorical data

Given: $x = (\text{Outlook} = \text{Sunny}, \text{Temperature} = \text{Cool}, \text{Humidity} = \text{High}, \text{Wind} = \text{Strong})$
 Predict: PlayTennis Yes or No?

Bayesian Rule	
$P(\text{Yes} x)$	0.0053
$P(\text{Yes} x) = \frac{[P(\text{Sunny} \text{Yes})P(\text{Cool} \text{Yes})P(\text{High} \text{Yes})P(\text{Strong} \text{Yes})]P(\text{Play}=\text{Yes})}{P(x)}$	
$P(\text{No} x)$	0.0206
$P(\text{No} x) = \frac{[P(\text{Sunny} \text{No})P(\text{Cool} \text{No})P(\text{High} \text{No})P(\text{Strong} \text{No})]P(\text{Play}=\text{No})}{P(x)}$	

Decision: Given the fact $P(\text{Yes}|x) < P(\text{No}|x)$, we decide x to be *No*.

$$P(\text{Outlook}=\text{Sunny}|\text{Play}=\text{Yes}) = 2/9$$

$$P(\text{Temperature}=\text{Cool}|\text{Play}=\text{Yes}) = 3/9$$

$$P(\text{Humidity}=\text{High}|\text{Play}=\text{Yes}) = 3/9$$

$$P(\text{Wind}=\text{Strong}|\text{Play}=\text{Yes}) = 3/9$$

$$P(\text{Play}=\text{Yes}) = 9/14$$

$$P(\text{Outlook}=\text{Sunny}|\text{Play}=\text{No}) = 3/5$$

$$P(\text{Temperature}=\text{Cool}|\text{Play}=\text{No}) = 1/5$$

$$P(\text{Humidity}=\text{High}|\text{Play}=\text{No}) = 4/5$$

$$P(\text{Wind}=\text{Strong}|\text{Play}=\text{No}) = 3/5$$

$$P(\text{Play}=\text{No}) = 5/14$$

Details,
refer to
next slide

Categorical data

Outlook	Play=Yes	Play=No
<i>Sunny</i>	2/9	3/5
<i>Overcast</i>	4/9	0/5
<i>Rain</i>	3/9	2/5

Temperature	Play=Yes	Play=No
<i>Hot</i>	2/9	2/5
<i>Mild</i>	4/9	2/5
<i>Cool</i>	3/9	1/5

Humidity	Play=Yes	Play=No
<i>High</i>	3/9	4/5
<i>Normal</i>	6/9	1/5

Wind	Play=Yes	Play=No
<i>Strong</i>	3/9	3/5
<i>Weak</i>	6/9	2/5

Total: $P(\text{Play} = \text{Yes}) = 9/14, P(\text{Play} = \text{No}) = 5/14$

Continuous data

For the training data

- Calculate the prior probability by counting the amount of data in each class. $P(\text{Window Glass})$ and $P(\text{Not Window Glass})$
- Build the likelihood probability distribution for each class, **assume** each attribute (column) follows a Gaussian distribution, and calculate its mean and variance.

For the test data

- Apply the same formula. Calculate the likelihood probability using the built Gaussian functions, $P(\text{Refractive index} = 1.51720 | \text{Window Glass})$ and $P(\text{Refractive index} = 1.51720 | \text{Not Window Glass})$

Training data	No.	Refractive index	Sodium	Magnesium	Aluminum	Silicon	Potassium	Calcium	Barium	Iron	Class
	1	1.51824	12.87	3.48	1.29	72.95	0.60	8.43	0.00	0.00	Window Glass
	2	1.51832	13.33	3.34	1.54	72.14	0.56	8.99	0.00	0.00	Window Glass
	3	1.51747	12.84	3.50	1.14	73.27	0.56	8.55	0.00	0.00	Window Glass
	4	1.51775	12.85	3.48	1.23	72.97	0.61	8.56	0.09	0.22	Window Glass
	5	1.51768	12.65	3.56	1.30	73.08	0.61	8.69	0.00	0.14	Window Glass
	6	1.51769	12.45	2.71	1.29	73.70	0.56	9.06	0.00	0.24	Window Glass
	7	1.51892	13.46	3.83	1.26	72.55	0.57	8.21	0.00	0.14	Window Glass
	8	1.51727	14.70	0.00	2.34	73.28	0.00	8.95	0.66	0.00	Not Window Glass
	9	1.51545	14.14	0.00	2.68	73.39	0.08	9.07	0.61	0.05	Not Window Glass
	10	1.51994	13.27	0.00	1.76	73.03	0.47	11.32	0.00	0.00	Not Window Glass
Test data	100	1.51720	13.38	3.50	1.15	72.85	0.50	8.43	0.00	0.00	



How to estimate distribution (easy data)?

- Problem: Given a collection of examples, estimate its distribution via **a parametric model** (e.g., Gaussian)
- Solution:
 - Assign a model to the distribution
 - Learn the parameters of the model from data

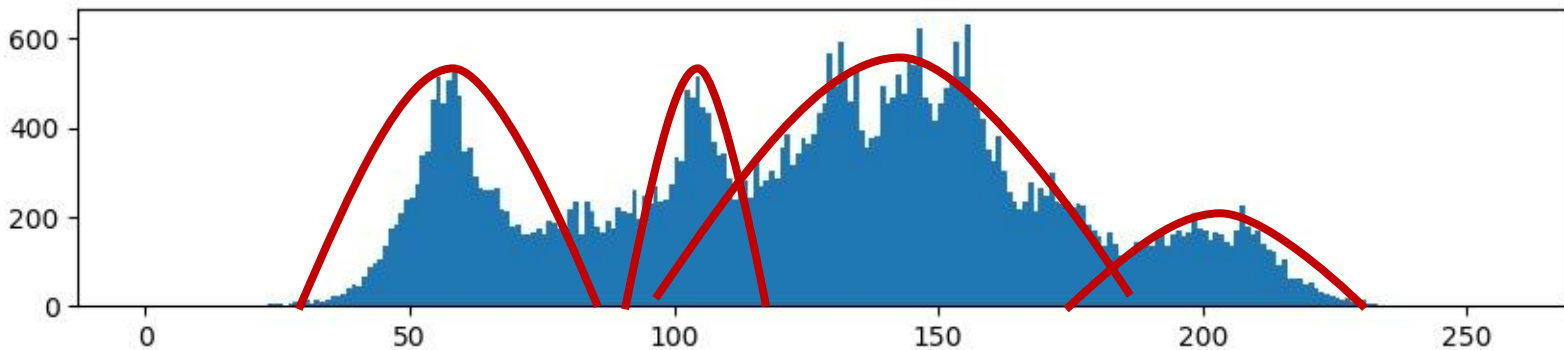
Gaussian distribution

$$p(\mathbf{x}|\boldsymbol{\mu}, \Sigma) = \frac{1}{\sqrt{(2\pi)^N |\Sigma|}} \exp \left\{ -\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}) \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right\}$$



How to estimate distribution (difficult data)?

- Example: The dataset (illustrated in the figure below) is complicated so that we can not use a **single** function (e.g., a single Gaussian function) to fit the data.
- We can fit the following dataset using **several** Gaussian functions (Gaussian mixture model). How do we know which data is used to fit which Gaussian function?
- We need to perform two tasks: data clustering (data labelling) and distribution estimation together.



Four Gaussian functions are used in this example.

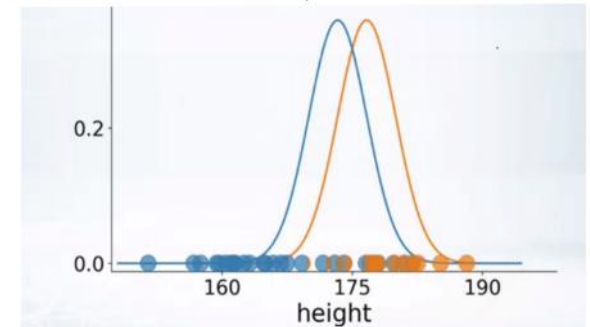
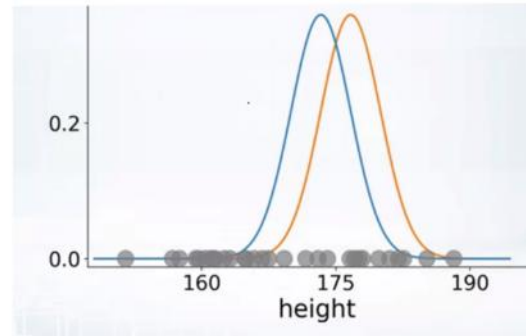
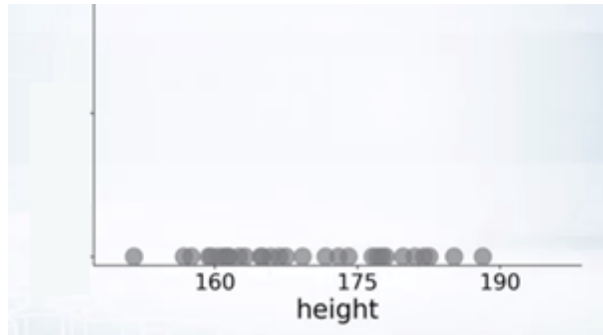


How to estimate distribution (difficult data)?

Dataset: Students' height
Objective: Estimate 2 clusters

Step 1: (randomly)
Initialize distribution

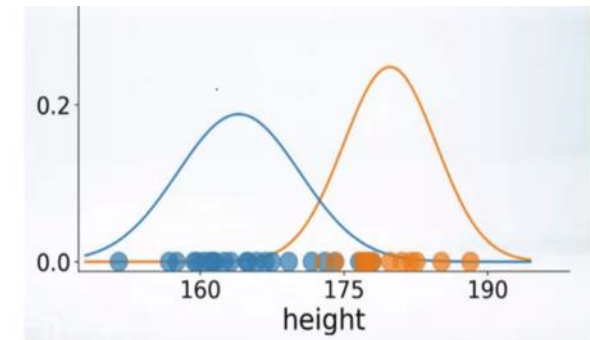
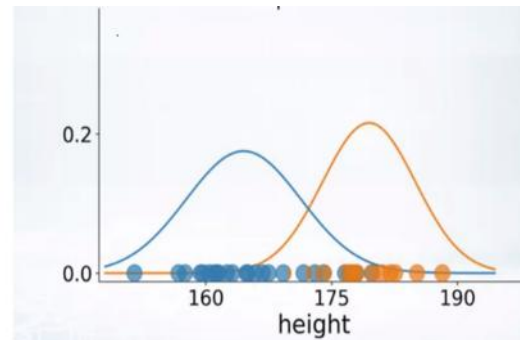
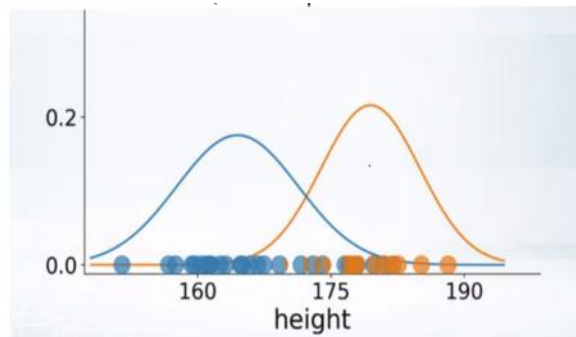
Step 2: Assign data label



Step 3: Update distribution

Step 4: Update data label

Step 5: Update distribution



Estimate distribution: A toy task

Example:

- A person rolls a single dice repeatedly
- You observe the set of outcomes
- You can estimate the distribution by counting
- $\text{Probability}(\text{number}) = \frac{\text{Count}(\text{number})}{\text{Sum}(\text{rolls})}$

Observed data

6	4	5	1	2	3	4	5	2	2	1	4	3	4	6	2	1	6
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---



Estimated distribution

X	1	2	3	4	5	6
$P(X)$	3/18	4/18	2/18	4/18	2/18	3/18

The probability $P(X) \neq \frac{1}{6}$ because it is estimated from the observed data.



Estimate distribution: A complicated task with additional rules

- Additional rules
 - Two dice (say, yellow dice and blue dice) are available.
 - There is a “caller” who randomly calls out the outcomes. At any time, you do not know which of the two dice you are calling out.
 - We observe the set of outcomes.
- How do you determine the probability distributions for the two dice? If you do not even know what fraction of time the blue numbers are called, and what fraction are yellow.



Observed data

6	4	5	1	2	3	4	5	2	2	1	4	3	4	6	2	1	6
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

Key idea: Introduce a hidden label

Objective: We need to formulate following distributions by filling these two tables.

X	1	2	3	4	5	6
$P(X \text{Blue})$						
$P(X \text{Yellow})$						

$P(Z = \text{Blue})$	
$P(Z = \text{Yellow})$	

- The caller will call out a number X IF
 - He selects “Yellow”, and the Yellow dice rolls the number XOR
 - He selects “Blue” and the Blue dice rolls the number X
- $P(X) = P(\text{Yellow})P(X|\text{Yellow}) + P(\text{Blue})P(X|\text{Blue})$
 - E.g. $P(6) = P(\text{Yellow})P(6|\text{Yellow}) + P(\text{Blue})P(6|\text{Blue})$

He selects Yellow

Dice rolls number 6

He selects Blue

Dice rolls number 6



Estimate distribution: Expectation maximization (EM)

- Iterative method
- Get some initial estimates for all parameters
 - Dice example: This includes probability distributions for dice AND the probability with which the caller selects the dice
- Two steps that are iterated:
 - **Expectation Step:** Estimate the values of unseen data labels
 - **Maximization Step:** Use the estimated values of the unseen data labels to estimate the model parameters

Expectation maximization

Step 1: Initialization

- We (guess) obtain an initial estimate for the probability distribution of the two sets of dice

X	1	2	3	4	5	6
$P(X \text{Blue})$	0.3	0.3	0.1	0.1	0.1	0.1
$P(X \text{Yellow})$	0.4	0.05	0.05	0.05	0.05	0.4

- We (guess) obtain an initial estimate for the probability with which the caller calls out the two shooters

$P(Z = \text{Blue})$	0.5
$P(Z = \text{Yellow})$	0.5

Expectation maximization

Step 2: Estimate hidden labels

- Every observed roll of the dice contributes to both “Yellow” and “Blue”

Recall that we randomly guess in Step 1

$$P(Z = \text{Blue}) = P(Z = \text{Yellow}) = 0.5$$

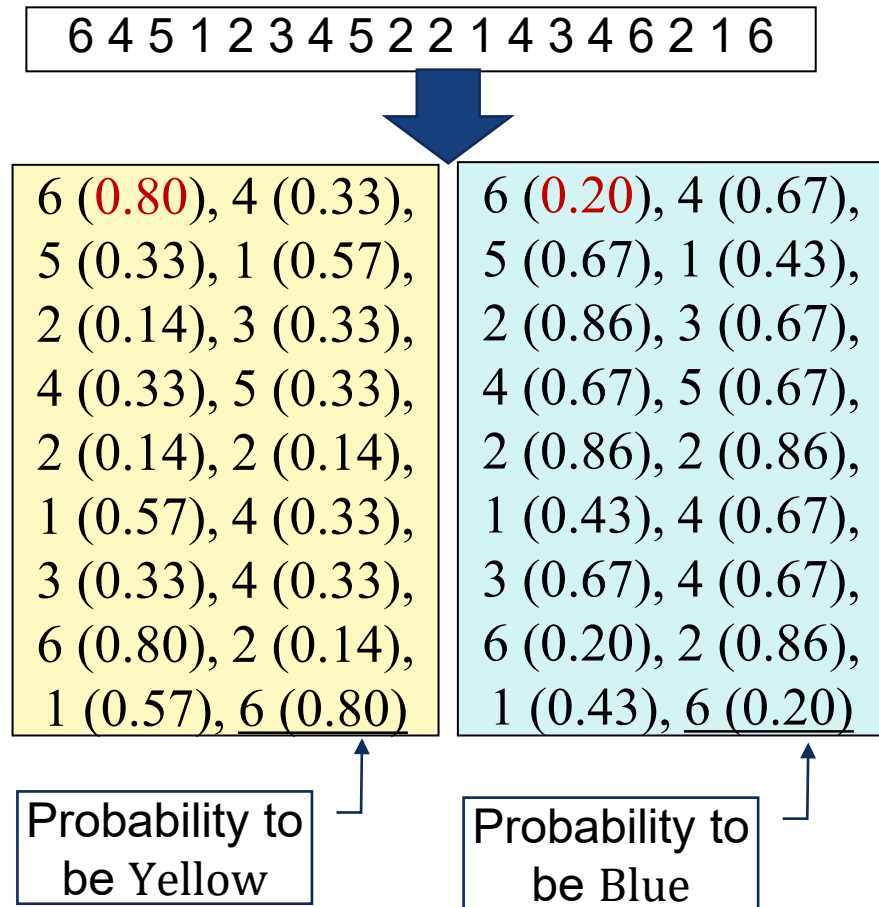
$$P(6|\text{Blue}) = 0.1, P(6|\text{Yellow}) = 0.4$$

For roll number 6, we estimate its label

$$P(\text{Yellow}|X = 6) = P(X = 6|Z = \text{Yellow})P(Z = \text{Yellow}) = 0.4 \times 0.5 = 0.2$$

$$P(\text{Blue}|X = 6) = P(X = 6|Z = \text{Blue})P(Z = \text{Blue}) = 0.1 \times 0.5 = 0.05$$

After normalization, we have $P(\text{Yellow}|X = 6) = 0.8$, $P(\text{Blue}|X = 6) = 0.2$



Expectation maximization

Step 2: Estimate hidden labels

- Every observed roll of the dice contributes to both “Yellow” and “Blue”
- Total count for “Yellow” is the sum of all the posterior probabilities in the Yellow column: 7.31
- Total count for “Blue” is the sum of all the posterior probabilities in the Blue column: 10.69

Called	$P(\text{Yellow} X)$	$P(\text{Blue} X)$
6	.80	.20
4	.33	.67
5	.33	.67
1	.57	.43
2	.14	.86
3	.33	.67
4	.33	.67
5	.33	.67
2	.14	.86
2	.14	.86
1	.57	.43
4	.33	.67
3	.33	.67
4	.33	.67
6	.80	.20
2	.14	.86
1	.57	.43
6	.80	.20
	SUM=7.31	SUM=10.69

Expectation maximization

Step 3: Update $P(X|\text{Yellow})$

- Total count for “Yellow”: 7.31
 - Total count for 1: 1.71
 - Total count for 2: 0.56
 - Total count for 3: 0.66
 - Total count for 4: 1.32
 - Total count for 5: 0.66
 - Total count for 6: 2.40
- **Update** probability of Yellow dice:
 - $P(1 | \text{Yellow}) = 1.71/7.31 = 0.234$
 - $P(2 | \text{Yellow}) = 0.56/7.31 = 0.077$
 - $P(3 | \text{Yellow}) = 0.66/7.31 = 0.090$
 - $P(4 | \text{Yellow}) = 1.32/7.31 = 0.181$
 - $P(5 | \text{Yellow}) = 0.66/7.31 = 0.090$
 - $P(6 | \text{Yellow}) = 2.40/7.31 = 0.328$

Called	$P(\text{Yellow} X)$	$P(\text{Blue} X)$
6	.80	.20
4	.33	.67
5	.33	.67
1	.57	.43
2	.14	.86
3	.33	.67
4	.33	.67
5	.33	.67
2	.14	.86
2	.14	.86
1	.57	.43
4	.33	.67
3	.33	.67
4	.33	.67
6	.80	.20
2	.14	.86
1	.57	.43
6	.80	.20
	SUM=7.31	SUM=10.69

Expectation maximization

Step 3: Update $P(X|\text{Blue})$

- Total count for “Blue”: 10.69
 - Total count for 1: 1.29
 - Total count for 2: 3.44
 - Total count for 3: 1.34
 - Total count for 4: 2.68
 - Total count for 5: 1.34
 - Total count for 6: 0.60
- **Update** probability of Blue dice:
 - $P(1 | \text{Blue}) = 1.29/11.69 = 0.122$
 - $P(2 | \text{Blue}) = 0.56/11.69 = 0.322$
 - $P(3 | \text{Blue}) = 0.66/11.69 = 0.125$
 - $P(4 | \text{Blue}) = 1.32/11.69 = 0.250$
 - $P(5 | \text{Blue}) = 0.66/11.69 = 0.125$
 - $P(6 | \text{Blue}) = 2.40/11.69 = 0.056$

Called	$P(\text{Yellow} X)$	$P(\text{Blue} X)$
6	.80	.20
4	.33	.67
5	.33	.67
1	.57	.43
2	.14	.86
3	.33	.67
4	.33	.67
5	.33	.67
2	.14	.86
2	.14	.86
1	.57	.43
4	.33	.67
3	.33	.67
4	.33	.67
6	.80	.20
2	.14	.86
1	.57	.43
6	.80	.20
	SUM=7.31	SUM=10.69

Expectation maximization

Step 4: Update $P(Z = \text{Blue})$ and $P(Z = \text{Yellow})$

- Total count for “Yellow”: 7.31
- Total count for “Blue”: 10.69
- Total instances: $7.31 + 10.69 = 18$
- We also normalize our estimate for the probability that the caller calls out Yellow or Blue
- $P(Z = \text{Yellow}) = 7.31/18 = 0.41$
- $P(Z = \text{Blue}) = 10.69/18 = 0.59$

Called	$P(\text{Yellow} X)$	$P(\text{Blue} X)$
6	.80	.20
4	.33	.67
5	.33	.67
1	.57	.43
2	.14	.86
3	.33	.67
4	.33	.67
5	.33	.67
2	.14	.86
2	.14	.86
1	.57	.43
4	.33	.67
3	.33	.67
4	.33	.67
6	.80	.20
2	.14	.86
1	.57	.43
6	.80	.20
	SUM=7.31	SUM=10.69



Expectation maximization: Summary

X	1	2	3	4	5	6
<u>Initially</u> , we randomly guess						
$P(X \text{Blue})$	0.3	0.3	0.1	0.1	0.1	0.1
$P(X \text{Yellow})$	0.4	0.05	0.05	0.05	0.05	0.4
Currently, we have updated results after <u>one iteration of update</u>						
$P(X \text{Blue})$	0.122	0.322	0.125	0.250	0.125	0.056
$P(X \text{Yellow})$	0.234	0.077	0.090	0.181	0.090	0.328

	Previous random guess	Updated results
$P(Z = \text{Blue})$	0.5	0.59
$P(Z = \text{Yellow})$	0.5	0.41

The *Expectation Maximization* (EM) algorithm will continue until the stopping criterion (e.g., # of iterations).

Thank you!

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